

PROJECT 10 REPORT

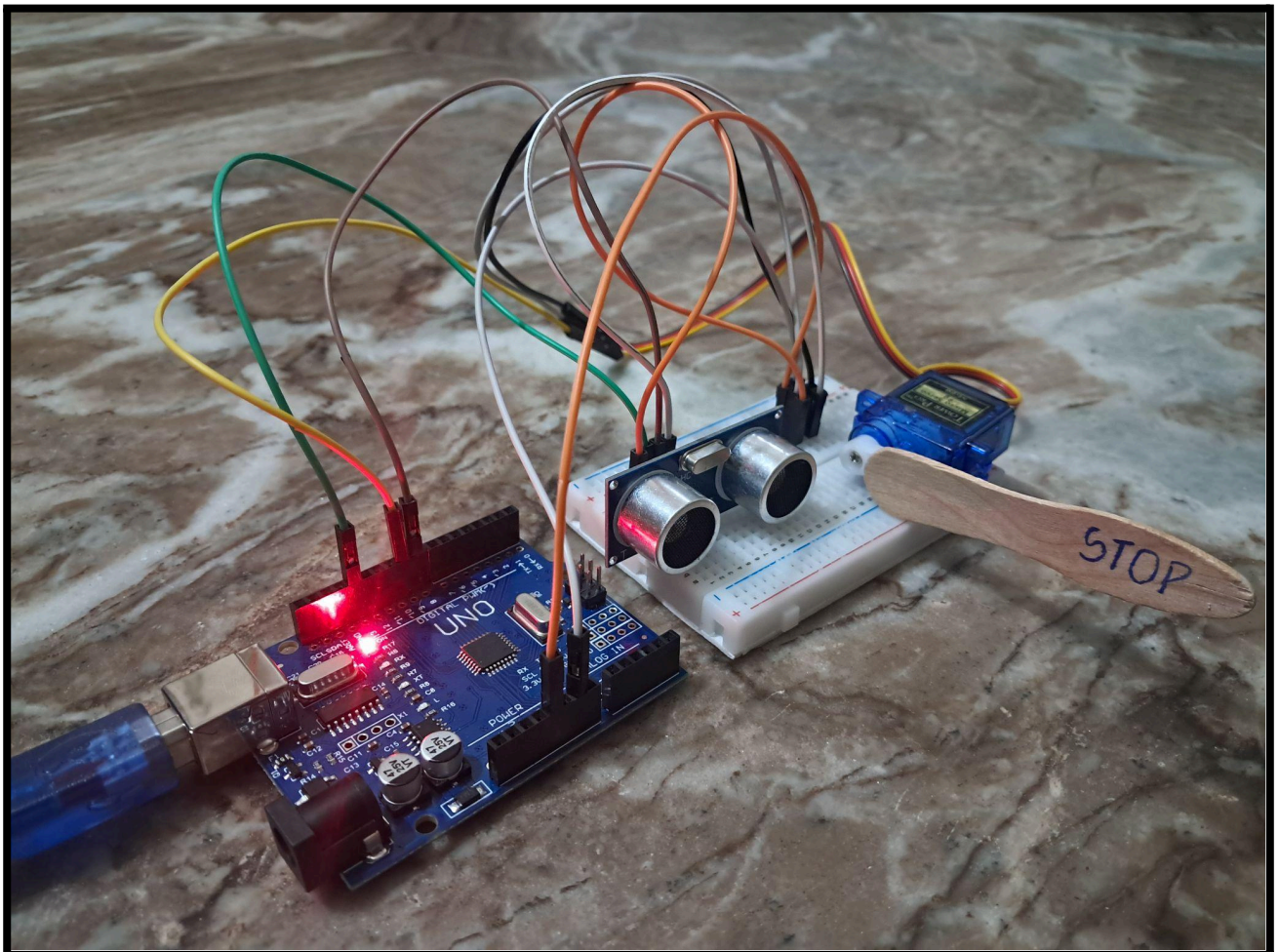
Automatic Parking Barrier System using HC-SR04 Ultrasonic Sensor and SG90 Servo Motor

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Duration : 1 day



1. Objective

The objective of this project was to develop an automatic parking barrier system using an HC-SR04 ultrasonic sensor and an SG90 servo motor.

The system automatically raises the parking barrier when a vehicle is detected near the entrance and lowers it after the vehicle passes creating a toll gate system.

2. Components Used

Component	Quantity
Arduino Uno R3	1
HC-SR04 Ultrasonic Sensor	1
SG90 Micro Servo Motor	1
Breadboard	1
Jumper Wires	Multiple
Ice Cream Stick (Barrier)	1
Double Sided Tape	As Required
USB Cable	1

3. Software Used

- Arduino IDE
- Servo Library (Servo.h)
- Windows Laptop

4. Theory

An automatic parking barrier is an embedded automation system that controls vehicle entry without human operation.

The HC-SR04 ultrasonic sensor continuously measures the distance of objects approaching the parking entrance by transmitting ultrasonic waves and measuring the time required for the reflected echo to return.

Arduino calculates the distance using:

$$\text{Distance} = (\text{Speed of Sound} \times \text{Time of Flight}) / 2$$

The calculated distance is compared with predefined threshold values.

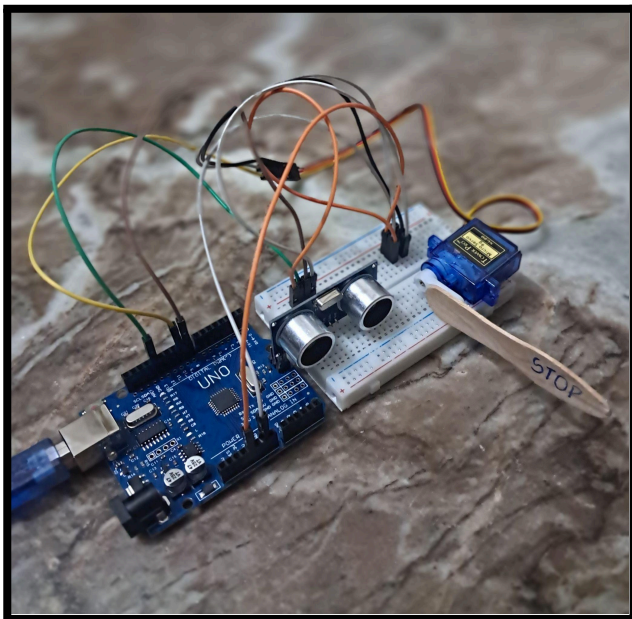
- **Distance < 15 cm** → The Arduino commands the servo motor to gradually rotate from 0° to 90°, raising the parking barrier.
- **Distance >= 15 cm** → Arduino rotates the servo 90° back to 0°, lowering the barrier.

Smooth movement was achieved using angle changes inside a for loop, producing realistic barrier motion.

State variables were introduced to remember whether the barrier was currently open or closed. This state based control improved system efficiency and prevented unnecessary repeated servo movements.

5. Tasks Performed

➤ Development of Automatic Parking Barrier System



Video

A complete automatic parking barrier prototype was developed using the HC-SR04 ultrasonic sensor and SG90 servo motor.

The ultrasonic sensor was made to continuously monitor the entrance for approaching vehicles.

Object Detected (<15 cm)

- Servo Rotates Smoothly
- Barrier Raises
- Vehicle Allowed to Pass

Object Removed (≥ 15 cm)

- Servo Rotates Back Smoothly
- Barrier Lowers

The automatic barrier was programmed to move gradually using for loops which produced realistic opening and closing motion.

State variables were implemented to store the previous state of the barrier.

The completed prototype successfully demonstrated automatic parking gate operation.

6. Learning Outcomes

During this project, the following concepts were learned:

- Automatic Parking Barrier Systems
- Sensor Actuator Integration
- Embedded Automation
- Ultrasonic Distance Measurement
- Servo Motor Position Control
- Smooth Servo Motion

- State Based Programming
- Previous State Memory
- Mechanical Barrier Design
- Prototype Development
- Practical Embedded System Debugging
- Mechanical and Electronic System Integration

7. Problems Faced

- I. The servo repeatedly executed the opening sequence while the vehicle remained in front of the sensor. (**State variables were introduced to remember whether the barrier was already open or closed**)

8. Observations

- The HC-SR04 accurately detected approaching objects.
- The servo successfully raised and lowered the parking barrier.
- Smooth servo motion produced more realistic operation than instant position changes.
- State based programming prevented unnecessary repeated barrier movement.
- Mechanical alignment of the servo and barrier arm significantly affected overall performance.
- The completed prototype successfully demonstrated an automated vehicle entry system.

9. Applications and Future Scope

The concepts learned in this project are used in Automatic Parking Systems, Toll Plaza Barriers, Industrial Entry Control and Automated Traffic Management Systems.

Possible future upgrade includes RFID Based Vehicle Authentication, Number Plate Recognition using AI, OLED Display for Gate Status and Multiple Vehicle Detection.

10. Conclusion

This project successfully demonstrated the development of an automatic parking barrier system using an HC-SR04 ultrasonic sensor and an SG90 servo motor.

The prototype provided valuable experience in developing reliable embedded automation systems.

This project represents the first **State-aware automation system** in the portfolio and strengthens the foundation for future robotics, industrial automation, and intelligent embedded control systems.